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The Impact of Pharmaceutical Care Intervention on the Quality of Life of Nigerian Patients Receiving Treatment for Type 2 Diabetes

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ABSTRACT

Objectives: To evaluate the impact of pharmaceutical care (PC) intervention on health-related quality of life (HRQOL) of patients with type 2 diabetes. **Methods:** This study was a randomized, controlled study with a 12-month patient follow-up. The study protocol was approved by the Research Ethical Committees of the institutions in which this study was conducted. A total of 110 patients were randomly assigned to each of the “intervention” (PC) and “control” (usual care [UC]) groups. Patients in the UC group received the usual/conventional care offered by the hospitals. Patients in the PC group received UC and additional PC for 12 months. The HUI23S4EN.40Q (developed by HUInc - Mark index 2&3) questionnaire was used to assess the HRQOL of the patients at baseline, 6 months, and 12 months. Two-sample comparisons were made by using Student's *t* tests for normally distributed variables or Mann-Whitney *U* tests for nonnormally distributed data at baseline, 6 months, and 12 months.

Comparisons of proportions were done by using the chi-square test. **Results:** The overall HRQOL (0.86 ± 0.12 vs. 0.64 ± 0.10 ; $P < 0.0001$) and single attributes except “hearing” functioning of the patients were significantly improved at 12 months in the PC intervention arm when compared with the UC arm. The HRQOL utility score was highly negatively (deficit $\geq 10\%$) associated with increasing age (≥ 52 years), diabetes duration (> 4 years), emergency room visits, comorbidity of hypertension, and stroke in both PC and UC groups. **Conclusion:** Addition of PC to UC improved the quality of life in patients with type 2 diabetes.

Keywords: HRQOL, patients with diabetes, pharmaceutical care intervention, quality of life, usual care.

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Introduction

Chronic medical conditions can impact multiple dimensions of health-related quality of life (HRQOL) [1]. Given that diabetes is part of a metabolic syndrome that increases the risk of heart disease and stroke [2], it is not uncommon for these conditions to occur as comorbidities in individuals with diabetes. Because comorbidities are prevalent in diabetes, it is unlikely that the HRQOL deficits associated with diabetes would be limited to the condition itself. Indeed, the presence and severity of complications or comorbidities have been associated with depression, anxiety, and impairment on multiple dimensions of HRQOL in diabetes [3]. The presence of cardiovascular complications as comorbidity with diabetes also leads to deficit in HRQOL [4].

The national standardized prevalence rate of diabetes mellitus in Nigeria is 2.2%, while the crude prevalence rate is 74% in those aged 45 years and above who live in urban areas [5]. Global estimates of the prevalence of diabetes for 2010 and 2030 showed that the prevalence of diabetes in Nigeria in 2010 was 4.7% and that it would be 5.5% in 2030 when compared with world population [6]. The complex nature of diabetes management prompted the Nigerian Ministry of Health to come up with a

standard treatment guideline to streamline the process of diabetes management and what service the patients should receive [7].

Several research studies have been carried out on health deficit associated with diabetes comorbidities. For instance, the work done by Maddigan et al. [8] to assess the impact of comorbid heart disease, stroke, and arthritis on HRQOL in people with diabetes in the general Canadian population concluded that “The illness burden experienced by individuals with diabetes is not only associated with diabetes itself, but largely with co-morbid medical conditions.” Also, Westaway [9] reported that chronic disease status and comorbidities were more important determinants of health and well-being than were ethnicity, age, language, gender, and marital status. Quality of life (QOL) is also increasingly recognized as an important health outcome in its own right, representing the ultimate goal of all health interventions [10].

The health utilities index Mark 3 (HUI3) classification system comprises eight attributes: vision, hearing, speech, ambulation, dexterity, emotion, cognition, and pain—each with five or six levels of ability/disability. Most of these attributes can be negatively affected by diabetes and its complications.

Pharmaceutical care (PC) is the direct, responsible provision of medication-related care with the purpose of achieving definite

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outcomes that improve a patient's QOL [10]. The principal elements of PC are that it is medication related; it is care that is directly provided to the patient; it is provided to produce definite outcomes; these outcomes are intended to improve the patient's QOL; and the provider accepts personal responsibility for the outcomes [10]. It is also the determination of the drug needs for a given individual and the provision of not only the required drug but also the necessary services (before, during, or after treatment) to ensure the optimally safe and effective drug therapy [11].

Diabetes is a disease that desperately needs more pharmacist involvement. Pharmacists who are specialized in this growing chronic condition can make a significant, positive impact on the patient, the health care system, and themselves [12]. Health care professionals are becoming increasingly aware of the need to assess and monitor the QOL as an important outcome of diabetes care. The QOL is an important outcome in its own right and because it may influence the patient's self-care activities, which may consequently have an impact on the diabetes control [13]. Many PC programs have been established in various countries to enhance clinical outcomes and the HRQOL. These programs were implemented by pharmacists, with the cooperation of physicians and other health care professionals. PC and the expanded role of pharmacists are associated with many positive diabetes-related outcomes, including improved clinical measures [14], improved patient and provider satisfaction [15,16], and improved cost of management [15,17]. The pharmacist can, therefore, in collaboration with physicians and other health care professionals, contribute to an improvement in the QOL of patients with diabetes by informing and educating patients, answering their questions, and, at the same time, monitoring the outcomes of their treatment [18]. Such interventions, however, are not very common in Nigeria. Therefore, this research was aimed at evaluating the impact of the PC intervention on the QOL of patients with type 2 diabetes mellitus in a tertiary hospital setting in Nigeria.

Methods

Research Design

This study was a randomized, controlled, and longitudinal prospective study with a 12-month patient follow-up. The study protocol was approved by the Research Ethical Committees of the University of Nigeria Teaching Hospital, Ituku/Ozalla, and Nnamdi Azikiwe University Teaching Hospital, Nnewi, in which this study was conducted. These hospitals are tertiary hospitals that serve as referral centers to most of the hospitals in Nigeria.

Patients with type 2 diabetes mellitus who fulfilled the entrance criteria were identified and included in the study. The inclusion criteria were as follows:

1. patients who were diagnosed with type 2 diabetes mellitus,
2. patients with type 2 diabetes who were receiving oral hypoglycemic and/or insulin therapy,
3. patients who provided written informed consent,
4. patients who expressed willingness to abide by the rules of the study, and
5. patients who were certified fit for the study by their consulting doctors.

Exclusion criteria were as follows:

1. patients who were diagnosed with type 1 diabetes (to avoid complexity in the study scope),
2. patients who were younger than 18 years (they are legally regarded as dependents and consequently they cannot take decisions of their own),

3. patients who were pregnant (they are generally not allowed to participate in a study of this nature by the institutions used for the study), and
4. patients who expressed willingness to withdraw from the study (participation is voluntary).

These criteria were according to the guiding principles of the institutional review boards of the hospitals used in this study. Following sample size determination, a sample size of at least 104 patients in each of the control and intervention groups was required. Based on these data, to ensure sufficient statistical power and to account for "dropouts" during the study, a target sample size of 220 patients was recruited (110 patients from each of the hospitals). The folders of the 110 selected patients in each hospital were assigned numbers 1 to 110, which represented an individual patient, and patients were randomly assigned to one of two groups (intervention group or control group) on the basis of the number on their folders by using online "random sequence generator" [19] with sequence boundaries of 1 to 110 (boundaries inclusive) set in a two-column format: the first column was priorly designated to the intervention group (55 patients) and the second column to the control group (55 patients).

Patients in the usual care (UC) group received the usual/conventional care offered by the hospitals, which included hospital visits on appointment or on a sick day, consultations with doctors, prescription of drugs and routine laboratory tests, review of diagnosis and medications, refilling of prescriptions by patients, and referral. This UC was offered with no education/training of the patients on their diseases and drugs and without empowerment of the patients to be fully involved in the self-management of their illnesses. Patients in the PC group received UC and PC for 12 months. This additional PC included a stepwise approach: setting priorities for patient care, assessing patients' specific educational needs and identification of drug-related problems, development of a comprehensive and achievable PC plan in collaboration with the patient and the doctor, implementation of this plan, and monitoring and review of the plan from time to time [10]. The nurses collaborated with the pharmacist in terms of organizing the patients and patients' folders, taking point-of-care testing, counseling the patients, and reinforcing the information given to the patients during training sections. The physicians provided the visitation/appointment schedule for the patients, and prescription of laboratory tests. They were also involved in the implementation of consensus strategies in managing drug-related problems in areas of changing, substitution, and withdrawal of medications.

The educational/training program for the patients consisted of four sessions of 90 to 120 minutes. The program covered the following areas: diabetes overview and its complications, self-monitoring blood glucose techniques and interpretation of diabetes-related tests, medications and their side effects, lifestyle modification, counseling, and effective interaction with health providers. PC provided ground for the patients to monitor and react to changes in their blood glucose levels, allowing them to integrate their diabetes into the lifestyle they preferred.

Data Collection

The HUI23S4EN.40Q (developed by HUInc - Mark index 2&3) questionnaire was used to assess the HRQOL of the patients. HUI23S4EN.40Q questionnaires were interviewer-administered to the patients in the intervention group and the control group at baseline, 6 months, and 12 months.

The HUI3 classification system comprises eight attributes. It defines 972,000 unique health states. Single-attribute scores of morbidity are defined on a scale such that the worst level has a score of 0.00 and the best level has a score of 1.00. Multiattribute

Table 1 – Baseline characteristics of the patients in PC and UC arms.

Demographic data	UC (n = 110)	PC (n = 110)	P
Mean age $\pm$ SD (y)	52.8 $\pm$ 8.2	52.4 $\pm$ 7.6	0.708
Grouped age: > 53 y, n (%)	81 (73.64)	75 (68.18)	0.373
Sex: Male, n (%)	49 (44.55)	44 (40)	0.495
Level of education, n (%)			0.406
Primary school	3 (2.72)	6 (5.45)	
Secondary school	71 (64.55)	63 (57.27)	
University	36 (32.73)	41 (37.27)	
Marital status, n (%)			0.409
Currently married	37 (33.64)	46 (41.82)	
Widowed	71 (64.54)	63 (57.27)	
Single	2 (1.82)	1 (0.91)	
Occupation, n (%)			0.611
Self-employed	37 (33.64)	34 (30.91)	
Employee	35 (31.82)	42 (38.18)	
Retired	38 (34.54)	34 (30.91)	
Smoking status: Smoker, n (%)	34 (30.91)	21 (19.09)	0.043*
Duration, mean $\pm$ SD	4.5 $\pm$ 2.2	4.8 $\pm$ 2.8	0.378
Duration: $\geq$ 5 y, n (%)	62 (56.36)	71 (64.55)	0.215
Family history of diabetes, n (%)	71 (64.55)	62 (56.36)	0.214
Physical activity/exercise, n (%)	18 (16.36)	23 (20.91)	0.387
Comorbidities, n (%)			
Hypertension	60 (54.55)	73 (66.36)	0.073
Congestive heart failure	11 (10.00)	15 (13.64)	0.404
Ischemic heart disease	7 (6.36)	8 (7.27)	0.789
Arthritis	37 (33.64)	43 (39.09)	0.400
$\geq$ 2 comorbidities	72 (65.45)	81 (73.64)	0.187
Overnight hospitalization, n (%)	9 (8.18)	7 (6.36)	0.604
Emergency room, n (%)	1 (0.91)	2 (1.82)	0.561
Use of insulin, n (%)	17 (15.45)	13 (11.82)	0.432
Oral antidiabetic medication, n (%)	103 (93.64)	107 (97.27)	0.1954
Other medications, n (%)			
Daily aspirin	43 (39.09)	57 (51.82)	0.058
Diuretics	71 (64.55)	84 (76.36)	0.055
Antihypertensives	98 (89.91)	78 (70.91)	0.0007*
Lipid-lowering	23 (20.91)	14 (12.73)	0.105
Complications, n (%)			
Myocardial infarction	2 (1.82)	4 (3.64)	0.408
Stroke	9 (8.18)	6 (5.45)	0.422
Foot ulcer	2 (1.82)	3 (2.73)	0.651
Blindness	1 (0.91)	1 (0.91)	1.000
Renal failure	3 (2.73)	8 (7.27)	0.122

PC, pharmaceutical care; UC, usual care.

* $P \leq 0.05$

utility functions convert comprehensive health state descriptions (i.e., vectors of one level for each attribute defined by a HUI classification system) into preference measures of overall HRQOL. The multiattribute scales of overall HRQOL are defined such that the score for dead is 0.00 and the score for perfect health is 1.00. Both HUI2 and HUI3 allow for negative scores of HRQOL that represent health states considered worse than dead. The lowest possible HRQOL scores are -0.03 for HUI2 and -0.36 for HUI3 [20]. Also collected were data on patients' demographics characteristics, lifestyles, and medical conditions, as outlined in Table 1.

Because we used two hospitals, we initially made comparisons of the groups (UC and PC) across the hospitals to determine their similarity, or, more specifically, to uncover any problems related to selection, history, or maturation effects. If the groups were found to be essentially similar in these respects, we planned to combine the groups for baseline, 6-month, and 12-month assessments of the effects of PC. If major differences were identified, we planned to analyze and report the group findings separately [21].

Data Analysis

Statistical analyses were performed by using the SPSS version 16. An intention-to-treat approach was used. Two-sample comparisons were made by using Student's *t* tests for normally distributed variables or Mann-Whitney *U* tests for nonnormally distributed data. Comparisons of proportions were done by using chi-square, Fisher's exact, or McNemar's tests. The differences in PC and UC were assessed at baseline, 6 months, and 12 months. An a priori significance level of *P* less than 0.05 was used throughout.

Results

The medical and educational content of the training course was rated positively by the 17 doctors and 29 nurses: the majority 38 (82.6%) rated the content as "excellent" and the remaining 8 rated the content as "very good" or "good"; only 3 (6.5%) of them

Table 2 – Changes in overall HRQOL and single attributes between the PC and UC groups after 12-mo follow-up.

Outcomes	Baseline			6th month		12th month		12-mo change		P	
	UC (n = 110)	PC (n = 110)	UC (n = 98)	PC (n = 102)	UC (n = 93)	PC (n = 99)	PC minus UC (95% CI)	Baseline	12 mo	6 mo	12 mo
Overall HRQOL	0.63 ± 0.04	0.61 ± 0.08	0.65 ± 0.05	0.79 ± 0.07	0.64 ± 0.10	0.86 ± 0.12	0.22 (0.1884–0.2516)	0.1029	<0.0001*	<0.0001*	<0.0001*
Vision	0.81 ± 0.10	0.77 ± 0.11	0.79 ± 0.09	0.94 ± 0.10	0.83 ± 0.10	0.97 ± 0.08	0.14 (0.1143–0.1657)	0.005*	<0.0001*	<0.0001*	<0.0001*
Hearing	0.90 ± 0.13	0.86 ± 0.21	0.91 ± 0.18	0.96 ± 0.19	0.92 ± 0.12	0.97 ± 0.23	0.05 (–0.0027 to 0.1027)	0.0908	0.0577	0.0577	0.0629
Speech	0.92 ± 0.21	0.89 ± 0.12	0.92 ± 0.09	0.94 ± 0.12	0.91 ± 0.08	0.95 ± 0.09	0.04 (0.0157–0.0643)	0.1947	0.1853	0.1853	0.0014*
Ambulation	0.82 ± 0.27	0.81 ± 0.19	0.83 ± 0.16	0.88 ± 0.29	0.84 ± 0.07	0.94 ± 0.10	0.10 (0.0753–0.1247)	0.7510	0.1526	0.1526	<0.0001*
Dexterity	0.79 ± 0.19	0.81 ± 0.12	0.84 ± 0.15	0.92 ± 0.09	0.80 ± 0.10	0.95 ± 0.22	0.15 (0.1008–0.1992)	0.3516	<0.0001*	<0.0001*	<0.0001*
Emotion	0.87 ± 0.12	0.85 ± 0.15	0.86 ± 0.20	0.90 ± 0.08	0.86 ± 0.15	0.96 ± 0.10	0.10 (0.0639–0.1361)	0.2760	0.0629	0.0629	<0.0001*
Cognition	0.89 ± 0.12	0.91 ± 0.18	0.88 ± 0.11	0.96 ± 0.14	0.85 ± 0.08	0.98 ± 0.12	0.13 (0.1008–0.1592)	0.3333	<0.0001*	<0.0001*	<0.0001*
Pain	0.67 ± 0.12	0.71 ± 0.10	0.70 ± 0.14	0.82 ± 0.15	0.68 ± 0.15	0.84 ± 0.10	0.16 (0.1239–0.1961)	0.008*	<0.0001*	<0.0001*	<0.0001*

CI, confidence interval; HRQOL, health-related quality of life; PC, pharmaceutical care; UC, usual care.

* $P \leq 0.05$.

suggested little modification or changes. With the exception of the number of participants taking hypertensive drugs and smoking, we found no other significant differences at baseline in both UC and PC groups. The number of patients who completed the study and whose data were analyzed at 6 months and 12 months in UC and PC groups were 98 (8.09%) versus 102 (92.73%) and 93 (84.55%) versus 99 (90.0%), respectively (Table 1).

The overall HRQOL of the patients was significantly improved at 6 months and 12 months in the PC arm when compared with the UC arm (0.79 ± 0.07 vs. 0.65 ± 0.05 ; $P < 0.0001$ and 0.86 ± 0.12 vs. 0.64 ± 0.10 ; $P < 0.0001$, respectively). The following single attributes were significantly improved in the PC arm over the UC arm at 6 months: vision (0.94 ± 0.10 vs. 0.79 ± 0.09 ; $P < 0.0001$), dexterity (0.92 ± 0.09 vs. 0.84 ± 0.15 ; $P < 0.0001$), cognition (0.96 ± 0.14 vs. 0.88 ± 0.11 ; $P < 0.0001$), and pain (0.82 ± 0.15 vs. 0.70 ± 0.14 ; $P < 0.0001$). There was no significant improvement in hearing, speech, ambulation, and emotion attributes. At 12 months, there were significant improvements in all the single attributes except hearing (Table 2). Increasing age (≥ 52 years, the overall mean age) had high negative impact (deficit $\geq 10\%$) on overall HRQOL in both PC and UC. A greater percentage of patients had ages greater than the overall mean age of about 52.6 years; this is reflected in more than one third of the patients being retirees. Patients older than 52 years, that is, older patients, had clinically significant lower QOL than did younger patients in both UC and PC groups. A majority of the patients had secondary education, and about a third were self-employed. Family history of diabetes was reported by a majority of the patients in both UC and PC groups. Those without a family history of diabetes had higher QOL in both groups. This result was consistent with what was expected, but it should be understood that some of the patients did not know their families' full medical history in detail. Diabetes duration (>4 years), emergency room visits, and comorbidity of hypertension and stroke had a high negative impact (deficit $\geq 10\%$) on the overall HRQOL in both PC and UC groups (Table 3). These factors were also associated with a low single-attribute score recorded in both arms of the study (Table 4). The changes in the overall HRQOL score and single-attribute scores of patients in the PC group were higher than in the UC group, which showed that the PC intervention had an overriding effect over the patients' characteristics than did UC. The percentage changes (improvements) in the PC group were higher than in the UC group (Table 3). Addition of PC to UC resulted in a significant gain in quality-adjusted life-years (QALYs) (PC vs. UC) (0.7625 ± 0.15 vs. 0.6425 ± 0.13 ; $P < 0.0001$), with 0.12 (0.07 to 0.1601; 95% confidence interval) QALY gained at the end of 12 months.

Discussion

Impact of the PC Intervention on the HUI3 Overall HRQOL

The overall HRQOL of the patients was significantly improved at 6 months and 12 months in the PC arm when compared with the UC arm. All the single attributes were significantly improved in the PC arm when compared with the UC arm at 12 months except hearing. The PC intervention impacted positively on patients assigned to it, and this change was both clinically (difference ≥ 0.03) and statistically significant [22,23]. PC interventions can have a positive change on the overall HRQOL and single attributes, that is, how patients with diabetes are able to cope with daily activities [24–26]. The improvement in the HRQOL may in part be attributed to the increased contact of patients with diabetes with the clinical pharmacist, but it is also likely to be associated with improved adherence to lifestyle advice.

The results of this intervention were similar to that of a prospective study on the impact of PC on QOL in patients with type 2 diabetes mellitus that was conducted in a private tertiary care teaching hospital in South India for a period of 8 months,

Table 3 – Impact of patients' characteristics on overall utility score of the patients.

Patients' characteristics	Overall HRQOL score after 12 mo (with characteristics minus without characteristics)				Change in overall HRQOL score after 12 mo (PC minus UC)			
	UC		PC		Deficit [*]		Change [†] (%)	
	With condition	Without condition	With condition	Without condition	UC	PC	UC	PC
Sociodemographic								
Age (≥52 y)	0.54	0.62	0.61	0.73	−0.08	−0.12	0.07 (13.0)	0.11 (17.7)
Family history of diabetes	0.59	0.65	0.67	0.72	−0.06	−0.05	0.08 (13.6)	0.07 (10.8)
Smoking status	0.58	0.62	0.64	0.72	−0.04	−0.08	0.06 (10.3)	0.10 (16.1)
Severity								
Use of insulin and oral medications	0.51	0.63 [‡]	0.56	0.74 [‡]	−0.12	−0.18	0.05 (9.80)	0.11 (17.5)
Use of oral medication only	0.58	0.63 [‡]	0.65	0.74 [‡]	−0.05	−0.09	0.07 (12.1)	0.11 (17.5)
Diabetes duration >4 y	0.56	0.62	0.67	0.74	−0.06	−0.07	0.11 (19.6)	0.12 (19.4)
Resource utilization								
Overnight hospitalization	0.58	0.60	0.62	0.73	−0.02	−0.11	0.04 (6.9)	0.13 (21.7)
Contact with physician in ER	0.59	0.61	0.68	0.76	−0.02	−0.08	0.09 (15.3)	0.15 (24.6)
Doctor visit more than 12 times	0.61	0.67	0.76	0.79	−0.06	−0.03	0.15 (24.6)	0.12 (17.9)
Comorbidity								
Hypertension	0.62	0.64	0.74	0.78	−0.02	−0.04	0.12 (19.4)	0.14 (21.9)
Stroke	0.42	0.59	0.57	0.82	−0.17	−0.25	0.15 (35.7)	0.23 (39.0)
Eye problems	0.63	0.66	0.69	0.73	−0.03	−0.04	0.06 (9.5)	0.07 (10.6)
Number of medical conditions ≥2	0.58	0.67	0.63	0.71	−0.09	−0.08	0.05 (8.6)	0.04 (6.0)

Note. Mean difference is the actual deficit associated with a particular condition (comorbidity, severity, and resource utilization). Negative values (−) indicate that the characteristic impacted negatively (lower utility scores) on the patients. ER, emergency room; PC, pharmaceutical care; UC, usual care.

^{*} Deficit, utility scores of patients with the characteristic minus utility scores of patients without the characteristics within the UC or PC group.

[†] Change, utility scores of patients in the PC group minus utility scores of patients in the UC group. "Deficit" and "change" values ≥0.03 or ≤0.03 are clinically significant.

[‡] Noninsulin nonantidiabetes medications (NINM).

Table 4 – Impact of patients' characteristics on single-attribute utility score for HUI3.

Single attributes	PC	Mean difference	PC	Mean difference	PC	Mean difference
Sociodemographic	Smoking status		Family history of diabetes		Age (≥ 52 y)	
Vision	0.8681	0.0707*	0.7977	0.0302	0.9295	0.1686*
Hearing	0.9013	−0.0380	0.9270	0.0063	0.9670	0.0534*
Speech	0.9962	0.0303	0.9671	0.0110	0.9890	0.0234
Ambulation	0.8928	0.0867*	0.8126	0.0257	0.9465	0.1752*
Dexterity	0.8189	0.0433	0.8198	0.0629*	0.8494	0.0928*
Emotion	0.9028	−0.0378	0.8746	0.0014	0.9154	0.0607*
Cognition	0.8157	0.0935*	0.7215	0.0420	0.8606	0.1699*
Pain	0.7096	0.1008*	0.6283	0.0077	0.7389	0.1554*
Resource utilization	Overnight hospitalization		Contact with physician in ER		Physician visit ≥ 12 visits	
Vision	0.8302	0.0315	0.8216	0.0445	0.8331	0.0487
Hearing	0.8522	0.0423	0.9378	0.0437	0.9471	0.0419
Speech	0.9861	0.0254	0.9825	0.0557*	0.9830	0.0249
Ambulation	0.8909	0.1251*	0.8688	0.2488*	0.8892	0.1541*
Dexterity	0.7996	0.0268	0.8199	0.2005*	0.8475	0.1948*
Emotion	0.8948	0.0407	0.8884	0.0855*	0.8648	−0.0217
Cognition	0.8035	0.1152*	0.7933	0.2886*	0.8254	0.199*
Pain	0.7059	0.1426*	0.6799	0.279*	0.7522	0.2923*
Severity	Use of insulin		Use of medication		Duration of diabetes > 4 y	
Vision	0.8319	0.0377	0.9750	0.1695*	0.8653	0.1278*
Hearing	0.9413	0.023	1.0000	0.0733*	0.8478	0.0436
Speech	0.9886	0.0328	1.0000	0.0284	0.9911	0.0448
Ambulation	0.9010	0.1568*	0.9670	0.1479*	0.9119	0.2098*
Dexterity	0.8531	0.1417*	0.9880	0.2128*	0.8403	0.1266*
Emotion	0.8858	0.0252	0.9280	0.057*	0.8956	0.055*
Cognition	0.8179	0.1552*	0.9760	0.244*	0.8424	0.2444*
Pain	0.7542	0.2563*	0.8610	0.2406*	0.7589	0.3029*
Comorbidity	Eye problem		Heart disease		Stroke	
Vision	0.9047	0.1909*	0.9190	0.1228*	0.8298	0.1091*
Hearing	0.9269	0.0074	0.9462	0.0185	0.9399	0.0654*
Speech	0.9914	0.0388	0.9300	0.0316	0.9849	0.0821*
Ambulation	0.9171	0.1908*	0.9190	0.1082*	0.8883	0.4262*
Dexterity	0.8790	0.1962*	0.8138	0.0327	0.8492	0.4371*
Emotion	0.9024	0.0601*	0.9521	0.0915*	0.8932	0.1332*
Cognition	0.8400	0.2017*	0.8352	0.1064*	0.7964	0.3598*
Pain	0.7591	0.2667*	0.7383	0.1238*	0.7158	0.5748*

Note. Mean difference represents the single-attribute utility score for patients in the PC group minus patients in the UC group. Positive value (+) indicates high single-attribute utility score for the PC group (i.e., HRQOL gained).

ER, emergency room; HRQOL, health-related quality of life; HUI3, health utilities index Mark 3; PC, pharmaceutical care; UC, usual care.

* Clinically significant for HUI3 single-attribute utility score (value ≥ 0.05).

which concluded that the PC program was effective in improving the clinical outcome and QOL of patients with type 2 diabetes mellitus [27]. Also, another 1-year observational study that evaluated the QOL in patients at the Medical University of South Carolina Family Medicine Clinic, who were followed by a clinical pharmacist diabetes educator, showed that patients rated their QOL high after the follow-up [28]. A 1-year study by Correr et al. [29] concluded that pharmacotherapy follow-up of patients with type 2 diabetes in community pharmacies can improve the HRQOL and satisfaction of patients.

Impact of Patients' Characteristics on Overall HRQOL

The health deficit imposed by old age cut across both groups, but the patients in the PC group has a significantly higher QOL than did those in the UC group, which showed that the intervention had an overriding effect over the effect of age on QOL. Low education and unemployment are associated with low income, which consequently affects the QOL of patients. This result is consistent with a study carried out in a Nigeria teaching hospital

by Issa and Baiyewu [30] who concluded that lower income, low education level, and low-rated employment affect the QOL of Nigerian patients with diabetes. The overall HRQOL scores of nonsmokers in both UC and PC groups were greater than scores of the smokers. This result was consistent with what was expected but could be attributed to the fact that the PC group had more female patients and there is sociocultural stigma associated with females smoking in Nigeria. The comorbidities considered in the study were eye problem, heart disease, and stroke. The respondents without eye problem had clinically significant higher QOL than did those with eye problem, in both UC and PC groups. This eye problem could be attributed to diabetes retinopathy, which is a complication of diabetes. This is consistent with a study performed in Canada [8] that concluded that the illness burden experienced by individuals with diabetes was associated not only with diabetes itself but largely with the comorbid medical condition. Lloyd et al. [31] concluded that the presence of even mild diabetes complications had a significant impact on patients' QOL. Early diagnosis and treatment is essential to help prevent the deterioration in the HRQOL of these patients.

A majority of the patients in this study had comorbidities. Stroke imposed the greatest HRQOL deficit among all other comorbidities in both PC and UC groups but more in the UC group. The HRQOL deficit associated with stroke and hypertension had been reported by Ekwunife et al. [32]. Their study demonstrated that patients with hypertension alone had a higher overall mean utility score while hypertensive patients with stroke had the lowest overall mean score. Other similar studies [2,9,33] revealed that stroke and other comorbidities can impose considerable health deficits on patients and concluded that the illness burden experienced by individuals with diabetes is associated not only with diabetes itself but largely with comorbid medical conditions. The results of this study also conform to a study that found that the presence of cardiovascular complications as comorbidity with diabetes also leads to deficit in HRQOL [3]. This emphasized the considerable public health impact that all these chronic conditions have on the HRQOL, particularly when they occur together.

This study has shown that diabetes complications have a profound effect on the HRQOL of patients with type 2 diabetes. Even the presence of mild diabetes complications has a significant impact on the HRQOL. In this study, the following four parameters were used to assess the severity of the disease: use of insulin and diabetes medications, use of diabetes medication only, duration of diabetes, and number of absenteeism from work or school in the last 1 year. Those who use a combination of insulin and oral antidiabetes medications and only oral antidiabetes medication had severe health deficits when compared with patients using noninsulin nonmedications in both UC and PC groups. This is consistent with the results of a study by Rubin [34] who assessed the feelings and complaints made by patients with diabetes using insulin and found that these could impact negatively on their QOL. This shows that as the disease progresses to a more severe situation, treatment option changes from diet to the use of oral medication only and deteriorates to the use of insulin in combination with oral medication.

This study revealed that those with a duration of diabetes of more than 4 years also had a considerable health deficit when compared with patients with a duration of diabetes of 4 years or less in both UC and PC groups. This result is consistent with a report from the American Diabetes Association [35] that stated that the longer the duration of diabetes the higher the chances of a patient developing overt nephropathy, which, in turn, lowers the HRQOL of the patients. To improve the HRQOL of patients with type 2 diabetes, early diagnosis of the disease and aggressive management of risk factors are necessary to prevent or delay the development of diabetes complications. Resource utilization was assessed as overnight hospitalization, contact with doctor in emergency room, and number of doctors' visit in the past 1 year. These resources included physicians, pharmacists, and nurses' time, hospital bed space, and spent more money. There was a considerable health deficit associated with resource utilization factors in both UC and PC groups. Patients who had overnight hospitalization and contact with physician or nurse in emergency room in the past 12 months had more health deficit when compared with patients who were not in their categories in both UC and PC groups.

Despite the enormous burdens imposed by the above-outlined patients' characteristics on the overall HRQOL, the patients in the PC group had significantly higher overall HRQOL scores than did their counterparts in the UC group, implying that the PC intervention had an overriding effect over the debilitating powers of the patients' characteristics.

Impact of Patients' Characteristics on the HUI3 Single Attributes

Comorbid stroke and old age (≥ 52 years) were associated with a relatively large mean difference between PC and UC groups in all single attributes. This indicates that patients in the PC intervention

who were 52 years or older and had stroke benefited most in all their single-attributes domains. Patients with diabetes in the UC group who were with comorbid medical conditions had lower utility scores on the anticipated diabetes-relevant single attributes than did their counterparts in the PC group, which resulted in a large mean difference. Based on the literature, it was anticipated that diabetes would affect the vision, dexterity, ambulation, emotion, and pain attributes of the HUI3 [4,36,37]. The considerable burden associated with diabetes and its comorbidities on these specific single attributes was therefore not surprising. Association of all sociodemographic, severity, comorbidity, and resource utilization factors, except family history, with the "pain" attribute was clinically and statistically significant. Ischemic heart disease, angina, and congestive heart failure are all associated with pain [38,39]. This is consistent with a study conducted by Ekwunife et al. [32] that demonstrated that the pain attribute had the lowest health state utility score among Nigerian patients with hypertension alone, hypertension and heart failure, hypertension and coronary heart disease, and hypertension and stroke. For the pain attribute in this study, the utility score recorded by the patients with stroke in the PC group far exceeded the score in the UC group after 12 months, resulting in a large mean difference. This means that the PC intervention was able to counter and override the utility-reducing power of comorbidities within 12 months. It is possible that some of the pain and discomfort (low score) found in the UC group was not attributable to stroke, but rather to another painful comorbid condition associated with diabetes and heart diseases, such as peripheral vascular disease [40,41].

Functioning on the "ambulation" attribute was similar to that of pain. All the sociodemographic, severity, comorbidity, and resource utilization factors were significantly associated with ambulation clinically. This means that the ambulation attribute of patients in the PC intervention group was highly improved compared with that of patients in the UC group. Also, the PC intervention had overriding benefits over all these factors that impair the QOL. The burden on the ambulation attribute associated with diabetes, comorbidities, and other factors was quite visible, not peculiar to one but all the factors. Clinically important burden on the emotion attribute was apparent for comorbidities and older patients. The largest emotional benefits of the PC intervention were observed in patients with diabetes with stroke compared with the UC group. These findings are consistent with a report of Maddigan et al. [8] that demonstrated that diabetes and its comorbidities could impact negatively on the overall utility state and single attributes.

QALYs associated with PC were significantly higher than those of UC. This indicates that extending this study beyond 1 year could offer more benefits to patients with diabetes in terms of QALYs gained. Some studies had demonstrated that the extension of PC beyond 1 year could offer extra benefits to patients with diabetes [42,43]. This study lends support to the use of the HUI3 for the evaluation of HRQOL as an outcome in PC intervention programs for patients with diabetes, as suggested by Feeny et al. [20].

Although the patients' characteristics affected the single attributes of the patients' HRQOL adversely, the intervention not only had overriding effects of these patients' characteristics that lower the overall HRQOL but also improved the single attributes of the QOL of these patients.

Limitations

Our study was subject to the following limitations, and the results were interpreted in this light: Selection bias was a threat because participation was voluntary though the groups were randomized. It remains possible that patients who chose to participate in the program may have differed in some important way from those who did not participate, which could pose a threat to external validity or generalizability. Given the difficulty in blinding in this

type of study, the clinicians involved in providing care and the interviewers administering the HUI3 questionnaire were not blinded to patient allocation. Recall bias was possible in this study because HUI3 was a 4-week recall questionnaire. Attrition bias or loss during follow-up was also a serious threat but was avoided by using an intention-to-treat design. The data were self-reported; however, self-reported data about diabetes status have been established to be both valid and reliable [44].

Conclusions

The addition of PC to UC resulted in improving the QOL of patients with type 2 diabetes. The overall HRQOL of the patients was significantly improved at 6 months and 12 months in the PC arm when compared with the UC arm. Vision, cognition, and pain single attributes were significantly improved in the PC arm over the UC arm after 6 months and all single attributes except hearing were significantly improved after 12 months. Stroke imposed the greatest HRQOL deficit among all other patients' characteristics in both PC and UC arms but more in the UC arm. The results of this study illustrate a convincing rationale for improving standards of care for patients with type 2 diabetes through the PC intervention. Further research, however, is needed to improve on the current PC intervention strategies such that the recorded improvements in HRQOL will be sustained for a very long time after an intervention.

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